

A Project Report
On
AirFly Insights

Submitted by

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Abstract

Airline delays, cancellations, and congestion present significant challenges to aviation operations and passenger experience. The **AirFly Insights** project delivers a comprehensive data-driven application designed to analyze, visualize, and interpret large-scale airline flight data, supporting informed decision-making for stakeholders. The system integrates historical flight records to examine key operational parameters such as delay causes, cancellation types, route congestion, and seasonal trends across multiple airlines and airports.

The project follows a milestone-based development approach encompassing data cleaning and feature engineering, exploratory data analysis, statistical visualization, and interactive dashboard development using Python, Pandas, NumPy, Matplotlib, Seaborn, and Streamlit. Advanced visualizations including bar charts, heatmaps, time series plots, and route-level analyses provide meaningful insights into airline performance and operational inefficiencies.

A dedicated interactive dashboard extends the analytical capabilities of the project by enabling dynamic exploration of airline performance, delay causes, and seasonal cancellation patterns. This dashboard supports comparative analysis across routes and airports, identifies delay-prone carriers, and highlights seasonal risks. The application is deployed locally via Streamlit, ensuring accessibility and user interaction.

Milestone 1: Data Foundation and Cleaning

Tasks Performed:

- Defined project goals, KPIs, and workflow for airline operations analysis.
- Loaded the Kaggle airline flight dataset using Pandas and explored schema, column types, dataset size, and null values.
- Performed sampling and memory optimization to handle large-scale data efficiently.
- Handled missing values in delay and cancellation columns.
- Engineered derived features including **Month, Day of Week, Hour, and Route (Origin → Destination)**.
- Formatted datetime columns for consistency and saved the cleaned dataset (cleaned_flights.csv) for reuse.

Insights Gained:

- The dataset contained significant missing values in delay and cancellation fields, requiring systematic preprocessing.
- Derived features enabled temporal and route-based analysis, improving the scope of later visualizations.
- Memory optimization and preprocessing ensured the dataset was manageable and consistent for large-scale analysis.

Outcome :

A cleaned and structured dataset with engineered features, ready for exploratory data analysis and visualization in subsequent milestones.

Milestone 2: Visual Exploration and Delay Trends

Tasks Performed:

- Conducted univariate and bivariate visual analysis of airlines, routes, and airports.
- Explored flight distribution across months, days of the week, and hours of the day.
 - Built visualizations including bar charts, histograms, boxplots, and line plots to examine operational patterns.
 - Compared delay causes (carrier, weather, NAS) across airlines and airports.
 - Visualized delay patterns by time of day and airport activity.

Insights Gained:

- Identified top airlines and busiest routes contributing to congestion.
- Flight activity showed clear peaks during specific hours and seasonal months.
- Carrier delays were the most frequent, while weather delays showed seasonal spikes.
- Certain airports consistently exhibited higher delay frequencies compared to others.

Outcome:

A comprehensive set of 8+ visualizations providing actionable insights into delay-prone airlines, peak delay periods, and airport activity patterns.

Milestone 3: Route, Cancellation, and Seasonal Insights

Tasks Performed:

- Identified the **top 10 origin–destination route pairs** to highlight the busiest flight corridors.
- Built **delay heatmaps** to visualize congestion patterns across airports and routes.
- Analyzed **monthly cancellation trends** and categorized cancellations by type (carrier, weather, NAS, security).
- Examined **seasonal patterns** such as winter months and holiday periods to assess their impact on flight reliability.
- Consolidated findings into visual summaries for stakeholder interpretation.

Insights Gained:

- Certain routes consistently showed higher congestion and delays, indicating operational bottlenecks.
- Winter months and holiday seasons exhibited elevated cancellation rates, confirming seasonal vulnerability.
- Carrier-related cancellations dominated overall, but weather-related cancellations spiked during specific months.
- Delay heatmaps revealed airports with persistent congestion, useful for prioritizing operational improvements.

Outcome:

Seasonal visual summaries and route-level insights were generated, providing a clear understanding of congestion hotspots, delay causes, and cancellation trends. These findings support data-driven decision-making for airline operators and analysts.

Milestone 4: Dashboard and Reporting

Tasks Performed:

- Designed and implemented an interactive **Streamlit dashboard** with multiple sections: Summary Dashboard, Data Overview, Traffic & Airport Activity, Route Analysis, Airport Performance, Delay Analysis, Delay Causes, Seasonal Insights, Airline Performance, and Delay Estimator.
- Integrated smart filters (Airline, Month, Origin Airport, Route) to allow dynamic exploration of data.
- Developed KPI metrics (total flights, average arrival delay, airports covered, cancelled flights) for quick insights.
- Created advanced visualizations including bar charts, pie charts, scatter plots, heatmaps, and bubble plots using Plotly Express and Plotly Graph Objects.
- Added a **Delay Estimator module** using a Linear Regression model to predict arrival delays based on departure delay, distance, and month.
- Documented the workflow with a user guide embedded in the dashboard for ease of navigation.

Insights Gained:

- The dashboard enables stakeholders to interactively explore airline performance, airport congestion, delay causes, and seasonal trends.
- Comparative analysis across airlines and airports highlighted delay-prone carriers and operationally stressed airports.
- Seasonal insights confirmed winter months and holiday periods as high-risk for cancellations and delays.
- The Delay Estimator provided predictive capability, demonstrating how machine learning can complement visualization.

Outcome:

A polished, interactive Streamlit dashboard (app.py) with integrated filters, KPIs, visualizations, and predictive modeling

Test Cases

Test Case ID	Scenario	Input	Expected Output	Result
TC01	Dataset not found	Remove cleaned_flights.csv from project folder	Error message: “✗ Dataset not found. Please make sure 'cleaned_flights.csv' is in the folder.”	Pass
TC02	Filter application	Select one airline from sidebar filter	Dashboard updates to show KPIs and charts for the selected airline only	Pass
TC03	Route analysis	Navigate to “✎ Route Analysis” section	Charts display top 10 routes by delay and flight count; bubble plot shows congestion intensity	Pass
TC04	Delay causes	Navigate to “⚠ Delay Causes” section	Bar and pie charts show distribution of delay causes (Carrier, Weather, NAS, Security, Late Aircraft)	Pass
TC05	Delay estimator	Enter Departure Delay = 20, Distance = 500, Month = 7	Predicted arrival delay displayed with gauge chart; value within realistic range	Pass

Errors Encountered and Resolutions

Error 1: Dataset not found

When the `cleaned_flights.csv` file was missing from the project folder, the dashboard displayed an error message and halted execution.

Resolution: Added a file existence check with a clear error message, ensuring users are guided to place the dataset correctly before proceeding.

Error 2: Empty filter selection

When no airline, month, or route was selected in the sidebar filters, the dataset returned empty results.

Resolution: Implemented default behavior to include all available values when no filter is applied, ensuring continuous visualization without interruption.

Error 3: Invalid filter combination

Selecting multiple filters that did not overlap (e.g., airline + route + month with no matching records) resulted in an empty dataset.

Resolution: Added a warning message “ *No data available for selected filters*” and stopped execution gracefully to avoid blank charts.

Error 4: Insufficient data for ML model

In the Delay Estimator module, when the dataset subset was too small or lacked variation, the model failed to train.

Resolution: Added safety checks for minimum data size and target variation, displaying user-friendly error messages instead of crashing.

Error 5: Extreme input values in prediction

Entering unrealistic values (e.g., very large negative departure delays) caused unstable predictions.

Resolution: Implemented bounds to cap predictions at a minimum of -60 minutes, ensuring outputs remain within realistic ranges.

Conclusion

The *AirFly Insights* project successfully demonstrates how raw airline operations data can be transformed into actionable insights through a structured, milestone-driven approach. Beginning with data cleaning and preprocessing, the workflow advanced through exploratory data analysis, statistical evaluation, and culminated in the development of an interactive Streamlit dashboard.

The dashboard integrates advanced visualizations — including bar charts, pie charts, scatter plots, heatmaps, and bubble plots — enabling intuitive exploration of flight activity, delay causes, congestion hotspots, and seasonal cancellation trends. The addition of smart filters ensures dynamic, user-driven analysis, while KPI metrics provide quick, high-level summaries of operational performance.

A key innovation is the **Delay Estimator module**, which applies machine learning to predict arrival delays based on departure delay, distance, and month. This predictive capability enhances the practical relevance of the system, allowing stakeholders to anticipate disruptions and plan accordingly.

Robust error handling was implemented to address challenges such as missing datasets, empty or invalid filter combinations, insufficient data for modeling, and extreme input values. These safeguards ensure stability, usability, and reliability across diverse scenarios.

Overall, *AirFly Insights* demonstrates technical rigor, professional presentation, and real-world applicability. It empowers airline operators, analysts, and decision-makers with interactive tools to assess performance bottlenecks, understand delay patterns, and support informed operational planning. The project stands as a comprehensive example of how **data visualization and predictive analytics** can be combined to deliver meaningful, scalable, and recruiter-ready solutions.

Future Enhancements

While the *AirFly Insights* dashboard provides robust visualizations and predictive insights, several enhancements can further increase its scalability, accuracy, and real-world applicability:

- **Integration of Live Flight Data:** Connect to airline APIs or aviation data providers to enable real-time monitoring of delays, cancellations, and route congestion.
- **Advanced Machine Learning Models:** Upgrade the Delay Estimator from Linear Regression to more sophisticated models such as Random Forest, Gradient Boosting, or Neural Networks to improve prediction accuracy.
- **Expanded Predictive Analytics:** Incorporate additional features such as weather conditions, airport traffic, and seasonal demand to provide more comprehensive delay forecasts.
- **Enhanced User Interface:** Improve dashboard aesthetics with modern UI components, customizable themes, and mobile-friendly layouts for broader accessibility.
- **Automated Reporting:** Add functionality to export insights as PDF or PowerPoint reports, enabling stakeholders to share findings quickly and professionally.
- **Scalability and Deployment:** Deploy the dashboard on cloud platforms (e.g., AWS, Azure, or GCP) to handle larger datasets and multiple concurrent users efficiently.
- **Role-Based Access:** Implement user authentication and role-based dashboards (e.g., airline operators, airport managers, analysts) for tailored insights.

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